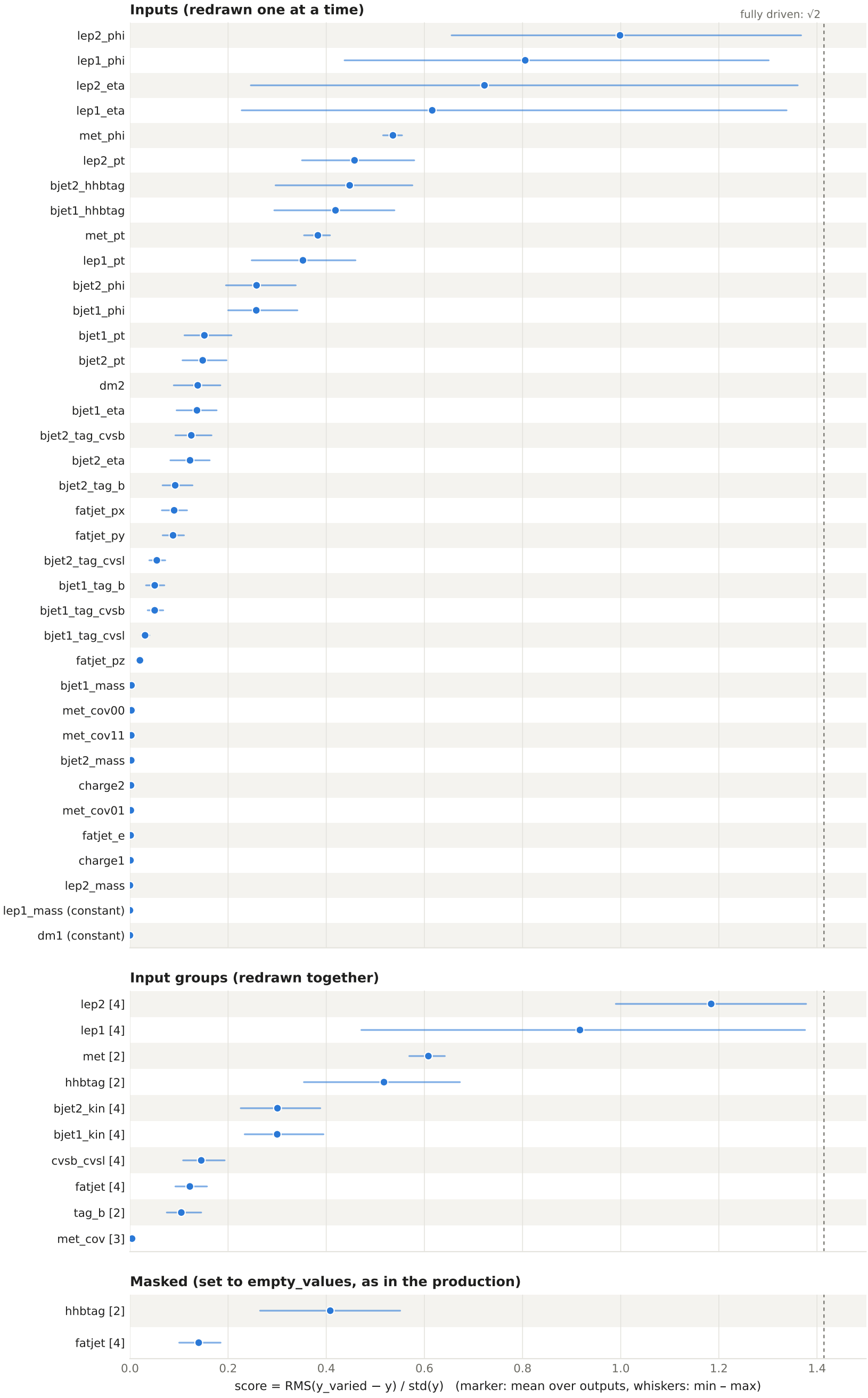
Sensitivity of uhh_nu_regression_friend (payloads/ml/uhh_nu_regression)

2018 diboson et



How the scores were made. N = 10000 (seed 0) random events are drawn from the input_ranges of the config 'uhh_nu_regression_friend' (build_config() called for 2018 diboson), every input independently, and evaluated with the config's own evaluate() (so derived features, rotations and the masking of empty values are exactly as in the production). For every row, the listed inputs are drawn again (or set to their empty_values for the masked rows) while all others keep their values, and the model is evaluated again. Score per output y: RMS(y_varied − y) / std(y). 0: the output does not change; √2 ≈ 1.41: the output changes as much as between two independent events, i.e. it is fully driven by these inputs. Marker: mean over the outputs (nu_px_1, nu_py_1, nu_pz_1, nu_px_2, nu_py_2, nu_pz_2); whiskers: smallest and largest single output. Masked rows show what replacing the inputs by their empty_values (as for inputs missing in the ntuples) changes. Caveat: the inputs are uncorrelated, so the scores show how the model responds over the given ranges (a wider range gives a larger score), not the importance on real events.